

For any democratic society to function, we need common ground¹ and deliberation² across groups with different values. Achieving this requires healthy information flows and exchanges³. Today we live in an information-rich world that has progressed from traditional newspapers, to the Web, and now Artificial Intelligence (AI)⁴. The advent of large language models⁵ powered conversational AI⁴ fundamentally shifts the human–information interface. Instead of navigating discrete webpages, users now engage in continuous, interactive dialogues and receive synthesized answers on demand. This paradigm promises unprecedented convenience, but how it transforms the ecology of information flows, and what consequences it holds for individuals and society, remains underexplored. My research bridges these gaps through an **interdisciplinary approach** that combines methods from Human–Computer Interaction (HCI), Natural Language Processing (NLP), **psychology**, and **political science**. I study the information flows with AI and how it impacts society. Furthermore, my research identifies new ways of Human-AI symbiosis that shape information flows to democratize and empower all humans by providing affordances to engage in equitable, informed and constructive information seeking, consumption and dissemination.

Overall, my research revolves around the key challenges to effective information exchange that have existed across changing information tools from word-of-mouth to web search and social media: **Information overload**⁶ makes it difficult to distill knowledge from data; **cultural and linguistic barriers**⁷ impede cross-community understanding; **echo chambers**⁸ and **filter bubbles**⁹ trap users in silos; **marginalized voices** go unheard in our current platforms because of imbalances of power dynamics in information exchanges¹⁰; and we lack robust mechanisms to assess the **credibility of information**, often forcing people to judge information quality based on group allegiance and spreading misinformation¹¹. These challenges **fuel opinion polarization, exacerbate information warfare, and hinder informed decision-making, ultimately threatening the stability of democracies**¹². To overcome these challenges, I organize my work into three complementary tracks: (1) Understanding and shaping human behavior during information seeking, consumption and dissemination with AI, (2) Understanding and shaping how AI operate in complex real-world contexts, to uncover and overcome barriers in global information access, and (3) Building tools and frameworks for democratized information governance. I aim to incorporate my research from these tracks to designing new interaction paradigms that empower users and democratize information exchange. Below are the details of each thrust.

Track I: Understanding and shaping Human Behavior with AI. Understanding and modeling human behavior allows us to design systems that elicit desired behaviors from users creating better information cycles. Classic theories exist for information seeking, such as **Information Foraging Theory** (IFT)¹³, but they were formulated for environments like the Web Search. However, AI introduces a fundamentally different setting: a single, continuous stream of information exchange. To motivate a new information foraging theory, we need to ask how this new human-information interface elicits behaviors that are different from web search. **In my study¹⁴, I show that this new interaction paradigm significantly amplifies confirmation bias and opinion polarization during information seeking.** I found, users interacting with a neutral LLM-based search assistant issued substantially more confirmatory queries (seeking information that supports their preexisting views) than those using a traditional web search. The conversational format encouraged them to ask more open-ended, subjective questions that inadvertently revealed their biases, which the neutral LLM then accommodated. Moreover, when the LLM was intentionally opinionated—configured to either agree with the user’s stance (consonant) or challenge it (dissonant)—the effects were starkly asymmetric. A consonant agent dramatically increased users’ confirmation bias and leading to opinion polarization, whereas a dissonant agent didn’t reduce bias. These findings uncovers new behaviors and challenges of information seeking with AI.

As a followup to these findings, **I am developing an IFT for Conversational Artificial Intelligence (IFT-AI)** to model human behavior. I plan to validate this framework with controlled experiments and looking at in-the-wild conversations, manipulating factors such as the agent’s response behaviour, various ui elements, the response format, etc. to observe the impact on user’s information seeking behavior. This is facilitated by a user study platform I developed, DeepForge, which is specifically designed to accelerate large scale understanding of user behavior while interacting with conversational agent. By integrating IFT-AI into DeepForge, researchers can how user behavior changes in response to scaffolds and affordances in conversational agents, promoting informed design choices. **Ultimately, this will provide a theoretical and empirical approach for studying user behavior and informing the design of safer, robust conversational search systems optimized for reducing bias and and fostering constructive information-seeking behaviors.**

Track II: Understanding and shaping Conversational Agent Behaviors Conversational Agents today are equipped with new scaffolds: Retrieval Augmented Generation¹⁵ allows models to retrieve articles and answer based on retrieved articles, Multilingual Understanding¹⁶ allows models to understand queries from different languages and synthesize articles from different languages. However, real-world contexts are pluralistic, often containing information conflicts; The narratives that the model outputs would have profound consequences on how users of Conversational Agents perceive the topic. To understand how Conversational Agents handle these pluralistic context, I ran an analysis¹⁷ of multilingual retrieval-augmented¹⁸ systems, **I found a strong monolingual bias**: for example, a query in a language α returned $\sim 68\%$ documents from the same language α

pool, and answers rarely combined sources from different languages (only 8.6% did). When information was unavailable in that language α , the model simply fell back to English, reinforcing high-resource language dominance instead of truly seeking and outputting cross-cultural information. This makes the current Conversational Agents propagators of linguistic filter bubbles hindering common ground.

To facilitate further research and development, *I am developing the Multilingual Narrative Benchmark, a systematic benchmark and dataset that evaluates AI's ability to handle conflicting narratives across different languages—particularly in sensitive geopolitical contexts, where objective ground truths are elusive and narrative differences are prominent.* This benchmark uses both synthetic and real world cases along with new metrics explicitly designed to assess whether models can overcome or inadvertently reinforce linguistic and cultural filter bubbles. Beyond text-based systems, my work expands into multimodal domains—particularly auditing Vision-Language Models (VLMs) similar to the Faux Polyglot audit of LLMs. Given the increasing use of VLM-generated images in newsrooms, social media, and meme cultures to propagate narratives, ensuring these models do not amplify polarization through culturally-biased imagery is critical.

Finally, to shape desired behavior Conversational Agents, we need to bring back humans in the AI pipeline. However, as current research has shown, high-quality human feedback is sparse which has motivated AI training pipelines to shift from human to AI feedback. This shift from human to AI feedback signals a troubling trend of human agnostic training pipelines. To counter this trend, we need to understand what prevents users from giving feedback and how we can design Conversational Agents to elicit scalable high-quality feedback. In our upcoming work at CHI 2026, Feedback by Design: Understanding and Overcoming User Feedback Barriers in Conversational Agents, we ran two studies to uncover the Feedback Barriers that prevent user feedback and show that following design desiderata that aim to follow cooperative principles of conversation elicit high-quality feedback. Ultimately, this approach seeks to create a sustainable, human-centric ecosystem where continuous user feedback shapes Conversational Agents.

Track III: Building tools and frameworks for democratized information governance Conversational Agents are one of the most powerful tools that we have had and the information disseminated and consumed through them would shape the society just how social media has done today. Although, the Calibrated trust in information is imperative for a healthy information economy and society¹⁹. To enable **diverse information seeking, calibrating trust and curb selective exposure & over reliance**²⁰ users need to be able to judge the credibility of information. The usefulness of an AI assistant in search is limited by whether users are able to trust information presented in a calibrated way. Two breakdowns are especially problematic: (a) source-based misinformation—the assistant uses misleading sources; and (b) hallucinated content—the assistant presents fabricated information. Furthermore, these sources can be consonant or dissonant to user beliefs. We rely on users to fact check these information but how everyday users fact check information is currently underexplored. To calculate this we use the information recall and reference clicks for information with different types of misinformation and biases. In a upcoming work, we found troubling trends where we saw users sparsely fact-checked information even when prompted to do so and we saw susceptibility to misinformation by labeling them as true regardless of whether that fact was fact-checked or not.

These insights inform my design of **Information Safety Protocol**, a conversational AI framework that calibrates how models linguistically present different types of information, clearly signaling credibility markers through UI to users thereby **decreasing cognitive load for verification**. By reducing the effort of verification and encouraging fact checking, these designs aim to *strengthen users' resilience to misinformation, simultaneously holding AI's accountable to adhering to the information safety standards.*

Conclusion. In summary, my research tackles the new ecology of information seeking in the generative AI era. This multi-faceted approach allows for diagnosis of problems like biased information foraging, cross-cultural filter bubbles, and lapses in information trust, but also to engineer concrete solutions. I envision conversational agents that actively counteract bias, bridge linguistic and community divides, support users in critical thinking, and facilitate inclusive knowledge exchange. By democratizing information access and putting human judgment at the center of the loop, such AI systems would serve as facilitators rather than gatekeepers of information. This has broad implications: enabling large-scale AI-mediated information exchange across communities, amplifying underrepresented voices, and reducing polarization by helping users find common ground. Moreover, by information safety protocols, my work empowers users to assess information quality for themselves.

Looking forward, my research aims to architect conversational systems that empower users to break down barriers to information access, consumption and proliferation. This involves creating a rich human centered information ecosystem with AI that enables democratic discourse, enables proliferation of all voices democratizing information dissemination, and guides users through complex conflicting narratives, fostering common ground.

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